Michael Remington Coulter

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Senior Software Engineer with nearly 8 years building engineering software and developer tools. Recent focus: coding-agent harnesses, MCP execution and evaluation.

PROFESSIONAL EXPERIENCE

Arm | Austin, TX

Senior Software Engineer — Apr 2023 - Present

Develop Interconnect Architect, an Angular/TypeScript SoC configuration tool with a Python backend. The product has about six years of customer releases and ships internally twice a month.

Recognition: Most Innovative Award, Engineering AI Showcase (Dec 2025): multi-agent system.

- Designed and built a shared coding-agent harness used by several Arm engineers; separated reusable tools/skills from project workflows and added source-verified knowledge capture.
  - Built `repair-jenkins`, a Codex skill teammates use to autonomously diagnose, fix and retest flaky cluster Playwright failures that otherwise take hours to days manually; requires three consecutive passes before manual patch review.
  - Developed an accompanying Rust/Tauri 2 workbench for repository/build/process control, with React/TypeScript UI, shared desktop/CLI core and bounded async Git inspection.
- Built `ia-mcp`, an internal Python Model Context Protocol (MCP) interface used by several engineers to translate interconnect intent into backend API calls, with typed plans, checkpoints, semantic validation and backend fallback.
  - Passed 92 contract and integration cases against the real Configuration Engine; separately tested Codex-driven NCI/CMN builds on cluster-installed modules and verified exact configuration-hash restoration in a rollback evaluation.
  - Built `ia-mcp` on Python 3.13; product-aware routing runs current backends directly and legacy Python 3.7 in an isolated, authenticated subprocess with exact-version checks.
- CMN GUI Lead: created and own all coherent-mesh GUI tooling from CMN-800 onward, including device/group configuration, custom routing and deadlock investigation.

Software Engineer — Apr 2020 - Apr 2023

- Built Flask services for Pelion's Jupyter notebooks, persisting files through Kubernetes storage backed by AWS EFS; designed PostgreSQL schemas and end-to-end tests.

Graduate Software Engineer — Jan 2019 - Apr 2020

- Built Verilog I/O test modules and C++ firmware for the Arm Mbed OS FPGA Test Shield.

TECHNICAL SKILLS

Languages: Python, Rust, TypeScript/JavaScript, C++, SQL
Agent systems: MCP, typed plans, semantic validation, checkpointing, rollback, knowledge provenance
Platform / delivery: Angular, React, Tauri 2, Flask, PostgreSQL; Kubernetes, Jenkins/LSF, CI/CD

EDUCATION

The University of Texas at Austin
Bachelor of Science, Electrical and Computer Engineering | Dec 2018
Primary focus: Computer Architecture and Embedded Systems | Secondary: Integrated Circuits